Analysis I

Uebungsblatt 10

Problem:	Comment:	Points:
1		26 / 36

1

(a)

Solution. We determine the closure $\bar{\mathbb{Q}}$. We claim that $\bar{\mathbb{Q}} = \mathbb{R}$.

In a normed vector space, the closure of a set A consists of all points x that are limits of sequences contained in A. Let $x \in \mathbb{R}$ be arbitrary. We rely on the density of rational numbers in $\mathbb{R}$. For any $n \in \mathbb{N} \setminus \{0\}$, there exists a rational number $q_n \in \mathbb{Q}$ such that:

$$|x - q_n| < \frac{1}{n}$$

Consider the sequence $(q_n)_{n\in\mathbb{N}}$. Since $\lim_{n\to\infty} \frac{1}{n} = 0$, convergence follows from the properties of the norm:

$$\lim_{n\to\infty} |x - q_n| = 0 \implies \lim_{n\to\infty} q_n = x$$

Since every $x \in \mathbb{R}$ can be written as the limit of a sequence in $\mathbb{Q}$, we have $\mathbb{R} \subseteq \bar{\mathbb{Q}}$. Since $\mathbb{Q} \subset \mathbb{R}$, it follows that $\bar{\mathbb{Q}} = \mathbb{R}$. □

(b)

Solution. We determine the interior $\mathring{\mathbb{Q}}$. We claim that $\mathring{\mathbb{Q}} = \emptyset$.

Recall the definition of an interior point in a normed vector space: A point $q \in \mathbb{Q}$ is an interior point if there exists an $\epsilon > 0$ such that the open ball $B_\epsilon(q) = \{y \in \mathbb{R} : |y - q| < \epsilon\}$ is essentially contained in $\mathbb{Q}$.

Assume for the sake of contradiction that $\mathring{\mathbb{Q}} \neq \emptyset$. Let $q \in \mathring{\mathbb{Q}}$. Then there exists $\epsilon > 0$ such that:

$$(q - \epsilon, q + \epsilon) \subseteq \mathbb{Q}$$

However, the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$. This implies that between any two real numbers $a < b$, there exists an irrational number. Therefore, in the interval $(q - \epsilon, q + \epsilon)$, there exists a number $\xi \in \mathbb{R} \setminus \mathbb{Q}$. This contradicts the assumption that the entire interval is contained in $\mathbb{Q}$.

Thus, no such interior point exists, and $\mathring{\mathbb{Q}} = \emptyset$. □

(c)

Solution. We determine the boundary $\partial\mathbb{Q}$. We claim that $\partial\mathbb{Q} = \mathbb{R}$.

The definition of the boundary of a subset A in a normed vector space is given by the closure minus the interior:

$$\partial A = \bar{A} \setminus \mathring{A}$$

Using the results from the previous parts:

- From part (a): $\bar{\mathbb{Q}} = \mathbb{R}$
- From part (b): $\mathring{\mathbb{Q}} = \emptyset$

9

Substituting these into the definition:

$$\partial\mathbb{Q} = \mathbb{R} \setminus \emptyset = \mathbb{R}$$

Thus, every real number is a boundary point of $\mathbb{Q}$. □

2

(a)

Solution. We aim to show that $f(x) = g(x)$ for all $x \in [a, b]$.

Let $x \in [a, b]$ be an arbitrary real number. Since the rational numbers $\mathbb{Q}$ are dense in $\mathbb{R}$, there exists a sequence of rational numbers $(q_n)_{n\in\mathbb{N}} \subset [a, b] \cap \mathbb{Q}$ such that:

$$\lim_{n\to\infty} q_n = x$$

By the hypothesis given in the problem, we know that f and g are equal on all rational numbers. Therefore, for every $n \in \mathbb{N}$:

$$f(q_n) = g(q_n)$$

We now take the limit as $n \to \infty$ on both sides. Since f and g are continuous functions on $[a, b]$, we can move the limit inside the function arguments (sequential continuity):

$$\begin{aligned} \lim_{n\to\infty} f(q_n) &= \lim_{n\to\infty} g(q_n) \\ f\left(\lim_{n\to\infty} q_n\right) &= g\left(\lim_{n\to\infty} q_n\right) \\ f(x) &= g(x) \end{aligned}$$

Since x was an arbitrary element of $[a, b]$, we conclude that $f(x) = g(x)$ for all $x \in [a, b]$. □

(b)

Solution. First, we determine the maximal domain D. The square root function is defined for non-negative real numbers. Thus, we require:

$$x^2 - 2x \geq 0$$

Factoring the expression gives:

$$x(x - 2) \geq 0$$

This inequality holds in two cases:

1. Both factors are non-negative: $x \geq 0$ and $x - 2 \geq 0 \implies x \geq 2$.
2. Both factors are non-positive: $x \leq 0$ and $x - 2 \leq 0 \implies x \leq 0$.

Therefore, the maximal domain is:

$$D = (-\infty, 0] \cup [2, \infty)$$

Next, we investigate the continuity at the boundary points $x = 0$ and $x = 2$. We treat f as a composition of continuous functions. Let $g(x) = x^2 - 2x$ and $h(y) = \sqrt{y}$. Then $f(x) = h(g(x))$. The polynomial g is continuous on $\mathbb{R}$, and the root function h is continuous on $[0, \infty)$.

At $x = 0$, we consider the left-sided limit (since the function is undefined for $x \in (0, 2)$):

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} \sqrt{x^2 - 2x} = \sqrt{0^2 - 0} = 0$$

The function value is $f(0) = \sqrt{0} = 0$. Since the limit equals the function value, f is continuous at $x = 0$.

At $x = 2$, we consider the right-sided limit:

$$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} \sqrt{x^2 - 2x} = \sqrt{4 - 4} = 0$$

The function value is $f(2) = \sqrt{0} = 0$. Since the limit equals the function value, f is continuous at $x = 2$. □

8

3

(a)

Solution. We analyze the continuity of f on the set $D = B(0, 1) \setminus \{0\}$.

For any $x \in D$, we have $x \neq 0$. Therefore, the function is strictly defined by the first case:

$$f(x) = \frac{x}{\|x\|_2}$$

We can view this as the product of a scalar function and a vector function:

$$f(x) = \left(\frac{1}{\|x\|_2}\right) \cdot x$$

1. The identity map $\mathrm{id} : x \mapsto x$ is continuous on $\mathbb{K}^m$. 2. The norm map $n : x \mapsto \|x\|_2$ is continuous on $\mathbb{K}^m$. 3. Since $x \neq 0$ for all $x \in D$, we have $\|x\|_2 \neq 0$. The reciprocal function $t \mapsto \frac{1}{t}$ is continuous on $\mathbb{R} \setminus \{0\}$.

Therefore, the term $\frac{1}{\|x\|_2}$ is a composition of continuous functions, making it continuous on D. Since f is the product of a continuous scalar function and a continuous vector function, f is continuous on D. □

(b)

Solution. We investigate the continuity of f on $B(0, 1)$. From part (a), we know f is continuous on $B(0, 1) \setminus \{0\}$. Therefore, we only need to check continuity at the point $x = 0$.

For f to be continuous at 0, we must have $\lim_{x\to 0} f(x) = f(0) = (1, 0, \ldots, 0)$. We test this using the sequential criterion.

Consider the sequence $(x_n)_{n\in\mathbb{N}}$ defined by approaching 0 from the negative direction of the first coordinate axis:

$$x_n = \left(-\frac{1}{n}, 0, \ldots, 0\right)^T$$

Clearly, $\lim_{n\to\infty} x_n = 0$. Now we evaluate the function values $f(x_n)$. Since $x_n \neq 0$, we use the first case of the definition:

$$\|x_n\|_2 = \sqrt{\left(-\frac{1}{n}\right)^2 + 0 + \ldots} = \frac{1}{n}$$

$$f(x_n) = \frac{x_n}{\|x_n\|_2} = \frac{(-1/n, 0, \ldots, 0)}{1/n} = (-1, 0, \ldots, 0)$$

The limit of this sequence of function values is:

$$\lim_{n\to\infty} f(x_n) = (-1, 0, \ldots, 0)$$

However, the function value at 0 is defined as $f(0) = (1, 0, \ldots, 0)$. Since $(-1, 0, \ldots, 0) \neq (1, 0, \ldots, 0)$ (assuming $1 \neq -1$ in $\mathbb{K}$), the limit does not match the function value.

Thus, f is not continuous at $x = 0$, and consequently not continuous on $B(0, 1)$. □

(c)

Solution. We investigate the continuity of the function $h : X \to \mathbb{R}$ defined by $h(x) := \|f(x)\|_2$. We calculate the value of $h(x)$ for the two cases defined in $f(x)$.

Case 1: $x \in B(0, 1) \setminus \{0\}$ For non-zero vectors, the function is defined as $f(x) = \frac{x}{\|x\|_2}$. Calculating the norm:

$$h(x) = \left\|\frac{x}{\|x\|_2}\right\|_2 = \frac{1}{\|x\|_2}\|x\|_2 = 1$$

Case 2: $x = 0$ For the zero vector, the function is defined as $f(0) = (1, 0, \ldots, 0)$. Calculating the norm:

$$h(0) = \|(1, 0, \ldots, 0)\|_2 = \sqrt{1^2 + 0^2 + \cdots + 0^2} = 1$$

Since $h(x) = 1$ for all $x \in B(0, 1)$, h is a constant function. Constant functions are continuous everywhere. Thus, h is continuous in $B(0, 1)$. □

9

top

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